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KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

SENIOR SCHOOL CURRICULUM DESIGN

ESSENTIAL MATHEMATICS

GRADE 10

2025

First Published in 2025

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FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory.

Towards achieving the mission of Basic Education, the Ministry of Education (MoE) embarked on a reform process that culminated in the adoption of a Competency Based Education (CBE) system in 2017. This came after extensive research, stakeholder engagement international benchmarking and consideration of the 21st Century socio-economic realities. CBE in Kenya adopted a 2-6-3-3 structure where the numbers indicate the years learners take at various levels in Basic Education (Pre-Primary, Primary, Junior School and Senior School; respectively). CBE has for critical elements: The Competency Based Curriculum (CBC), Pedagogy (teaching and learning), Resources and the Competency Based Assessment. The MoE oversees the provision of resources while the other components (Curriculum, Pedagogy and Assessment are handled by the Kenya Institute of Curriculum Development, (KICD), Teachers Service Commission, (TSC) and the Kenya National Examinations Council (KNEC); respectively.

KICD has successfully developed curriculum designs that have been used in the implementation of the Competency Based Curriculum (CBC) for the Pre-Primary, Primary and Juniors School levels. The first cohort of learners pursuing the CBC will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level are expected to facilitate learners to acquire relevant knowledge, skills, values and attitudes (competencies) in preparation for tertiary education and the world of work. I urge all agencies of the GoK and other stakeholders in Education to use the designs to plan for effective and efficient implementation of Competency Based Education.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Education (CBE) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The Kenya Institute of Curriculum Development (KICD) developed Competency Based Curriculum (CBC) designs that were used for pilot studies in 2017 and 2018 before the roll-out at Pre-Primary and Primary Early Year (Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through *a phase in –phase out* strategy, to ensure a smooth transition. Currently, the first cohort of CBC learners are at Grade 9 – the last grade at Junior School. The implementation of CBE has since benefited immensely from the recommendations of the *Presidential Working Party on Education Reforms* (PWPER) that were released in August, 2023 and fully implemented, in terms of curriculum designs.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways, tracks and subjects of their choices guided by the potentials demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and training and/or initial competencies for the Industry. This is an important step in the realisation of the Vision and Mission of the curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They also present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

The Ministry of Education, through the State Department for Basic Education commits to provide resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula, programmes and curriculum support materials for Basic and Tertiary Education and Training. The curriculum development process involves research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017), that responds to the demands of the 21st Century, the United Nations Sustainable Development Goals (SDGs). African Agenda 2063, East African Community Protocol and the aspirations captured in the Kenya Constitution 2010 and the Kenya Vision 2030.

KICD receives its funding from the GoK to enable the successful achievement of the stipulated mandate and implementation of the Government and Education sector plans. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank and GoK through the *Kenya Primary Education Equity in Learning Programme* (KPEELP); a project coordinated by the MoE. Therefore, the Institute is very grateful to the GoK, through the MoE and the development partners (especially The World Bank) for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also wish to acknowledge the KICD curriculum development officers and other staff, all teachers, educators who took part as panellists; the institutions that participated and representatives of various stakeholders for their roles in the development of the Grade 10 curriculum designs. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson and other members of the Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBE at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.
2. **Promote the social, economic, technological and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
   Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.
   b) **Economic Needs**
   Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy which is in need of an adequate and relevant domestic workforce.
   c) **Technological and Industrial Needs**
   Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.
3. **Promote individual development and self-fulfilment**
   Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

4. **Promote sound moral and religious values.**
   Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.
5. **Promote social equity and responsibility.**
   Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.
6. **Promote respect for and development of Kenya's rich and varied cultures.**
   Education should instil in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.
7. **Promote international consciousness and foster positive attitudes towards other nations.**
   Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.
8. **Promote positive attitudes towards good health and environmental protection.**
   Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/ pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be ***engaged, empowered and ethical citizens*** ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas** namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
4. Exploit individual talents for leisure, self-fulfilment, career growth, further education and training.
5. Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Wood Technology 36. Media Technology 37. Marine and Fisheries Technology

Note:

a) Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
b) Every school shall offer Pastoral Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.
c) There shall be One lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

Table 2, below provides guidelines on distribution of lessons at Senior School.

LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
** Physical Education*	*3*
** ICT Skills*	*2*
**PPI*	*1*
** Learner Personal /Group Study*	*1*
Total Number of Lessons	**40**

ESSENCE STATEMENT

Essential Mathematics in Senior School builds on and deepens the competencies acquired by the learner from Junior school. It plays a pivotal role in developing the learner's computational abilities as well as critical thinking and problem solving skills. Specifically, the subject focuses on numbers and algebra as foundation to the course in developing numerical and algebraic thinking. It also includes measurements of different quantities, particularly commercial arithmetic which is important for acquisition and application of financial literacy in real life. Further, some concepts in geometry are included since they are useful in spatial relations as well as navigational skills while Statistics and Probability are critical in research and in understanding different population dynamics and phenomena that are useful in solving problems in day to day life.

This Mathematics curriculum has explicitly suggested learning activities that focus on the mathematical process skills of problem solving, communication, connection, collaboration and reasoning as core to acquisition of the desirable competencies. Similarly, for holistic development and wellbeing of learners Pertinent and Contemporary Issues such as Life Skills, Citizenship and Education for Sustainable Development as well as Values are integrated into this Mathematics course.

Essential Mathematics is crucial in supporting and enhancing acquisition of knowledge and skills in other subjects. Further, it is critical in determining career choices in the fields of Social Sciences, Arts, Sports, vocational and humanities. Careers in areas such as media, hospitality industry, guidance and counseling, sole proprietorship, sports management, coaching, refereeing, journalism and photography will benefit from this Mathematics. More importantly, its aesthetic and utility value in day to day activities will contribute to an informed numerate citizenry.

SUBJECT GENERAL LEARNING OUTCOMES

By the end of the course the learner should be able to:

1. Acquire numerical computational skills, concepts and competencies for continuous learning and application of mathematical skills in real life,
2. Apply measurement skills of different quantities used in daily activities,
3. Evaluate and make financial decisions and transactions in real-life situations,
4. Collect, analyse and represent data for informed decision-making in real-life situations,
5. Use mathematical concepts and skills to support other learning subjects for further training, research and career development,
6. Develop confidence and interest in mathematics as well as the coherence, utility and aesthetic nature of the subject for lifelong learning,
7. Apply critical and creative thinking, logical reasoning and analytical skills for problem-solving,
8. Use digital technology in computations, explorations, innovations, navigational skills and for further learning,
9. Demonstrate appropriate responsiveness to Pertinent and Contemporary Issues in the community and promote values for peaceful coexistence.

STRANDS

The Essential Mathematics curriculum has 3 strands namely:

1. Numbers and Algebra,
2. Measurements and Geometry, and
3. Statistics and Probability.

SUMMARY OF STRANDS AND SUB STRANDS

S/No.	Strand	Sub strand
1.	Numbers and algebra	1.1 Real Numbers
		1.2 Indices
		1.3 Quadratic Equations
2.	Measurements and Geometry	2.1 Similarity and Enlargement
		2.2 Reflection
		2.3 Trigonometry
		2.4 Area of Polygons
		2.5 Area of part of a circle
		2.6 Surface area of solids
		2.7 Volume and Capacity
		2.8 Commercial Arithmetic 1
3	Statistics and Probability	3.1 Statistics 1
		3.2 Probability 1

STRAND 1.0: NUMBERS AND ALGEBRA

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers and Algebra	**1.1. Real Numbers** • *Operations on rational numbers* • *Reciprocal of numbers* • *Application of Integers*	By the end of the sub strand, the learner should be able to: a) classify real numbers as rational or irrational, b) perform combined operations on rational numbers, c) determine the reciprocal of real numbers using calculators, d) apply rational numbers in real life situations, e) appreciate the use of real numbers in real life.	The learner is guided to: • brainstorm and estimate individual weight and height, check shoe size and record on a table. Classify the numbers as odd, even, prime and composite (learners to observe emotional issues when estimating weight and height). • discuss with peers the real numbers and classify them as either rational or irrational numbers. • read and interpret case scenarios involving rational numbers and work out the combined operations. • discuss situations involving combined operations such as distribution of learning materials and work out relevant operations, • use a calculator to work out the reciprocal of real numbers • discuss how real numbers are used real life situations such as body temperature and blood pressure.	Why are numbers important?

Core competencies to be developed:

- Critical thinking and Problem solving: the learner develops interpretation and inference skills as they classify numbers as either rational or irrational numbers.
- Self-efficacy: the learner develops self-awareness as they discuss the use real numbers in real life such as in body temperature and blood pressure.

Values:

- Responsibility: the learner takes care of a calculator when using it to work out reciprocal of numbers.
- Respect: the learner develops humility by taking turns while discussing with peers and working out combined operations.

Pertinent and Contemporary Issues (PCIs):

Health promotion issues: the learner develops healthy practice skills while discussing use of real numbers in recording body temperature and blood pressure.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers and Algebra	**1.2. Indices** • *Indices (powers and bases)* • *Laws of Indices* $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$ $a^0 = 1$ where *m* and *n* are whole numbers Applications of indices	By the end of the sub strand, the learner should be able to: a) express numbers in index form in different situations, b) deduce the four laws of indices from powers and bases, c) apply the laws of indices in numerical computations, d) recognise the use of indices in computations.	The learner is guided to: • Read the numbers of their height or weight and discuss how such numbers can be written in terms of their prime factors to generate bases and powers, • search for information on numbers such as population of an area, write the number to the nearest thousands then express it to index form, • write index numbers in the expanded form and extract the of laws of indices, • use the laws of indices to simplify the powers of numbers with the same base, • work with peers to use the laws of indices in working out mathematical computations, • use digital resources or print resources to learn more on laws of indices and their applications in real life situations.	1. Why are indices important? 2. How are the laws of indices applied in real life?

Core competencies to be developed:

- Learning to learn: the learner shares learnt knowledge as they discuss with peers and use the laws of indices to work out mathematical computations.
- Digital literacy: the learner develops the skills of connecting and interacting with digital technology when using digital resources to search for the laws of indices.

Values:

- Unity: the learner develops fairness and team spirit as they carry out given mathematical tasks involving the laws of indices
- Responsibility: the learner develops leadership and guidance while discussing with peers as they work on the group tasks.

Pertinent and Contemporary Issues (PCIs):

Social cohesion: the learner works with peers harmoniously to carry out given mathematical tasks involving laws of indices

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Numbers and Algebra	**1.3 Quadratic Equations (I)** • *Formation of quadratic expressions* • *Factorisation and solve quadratic equations* • *Applications of quadratic equation*	By the end of the sub strand, the learner should be able to: a) form algebraic expressions from different situations, b) form quadratic expressions and equations from real life situations, c) factorise quadratic expressions in different situations, d) solve quadratic equations by factorisation, e) recognise use of quadratic expressions in real life situations.	The learner is guided to: • read and interpret relevant case scenarios and use letters to represent unknown quantities forming algebraic expressions, • read and interpret algebraic statements that relates to situations such as areas of rectangle of unknown sides, number of chairs arranged in rows and columns to generate quadratic equations, • discuss with peers and form quadratic expressions and equations from the statements • discuss with peers and work out factors of quadratic expressions such as the product and sum of number of seedlings planted in rows and columns in a rectangular farm of known area, • work out solution of quadratic equations by factor method, • use digital resources or print media to search on the application of quadratic equations, and share with peers. Observe online safety measures.	How are quadratic equations applied in real life?

Core competencies to be developed:

- Communication and collaboration: the learner collaborate with peers as they discuss with peers and work out factors of quadratic expressions.
- Critical thinking and Problem solving: the learner develops open-mindedness, discuss with peers and works out factors of quadratic expressions.

Values:

Respect: the learner displays humility by listening to others opinions while discussing areas where quadratic equations are used in real life situations.

PCIs:

Online safety: the learner observes online safety procedures and guidelines while searching on the application of quadratic equations using digital resources.

Suggested Assessment Rubric

Level / Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to classify real numbers as rational or irrational and perform combined operations on rational numbers	The learner correctly classifies real numbers and performs combined operations on rational numbers with varied examples.	The learner correctly classifies real numbers and performs combined operations on rational numbers	The learner either classifies real numbers or performs combined operations on rational numbers with minor errors	The learner either classifies real numbers or performs combined operations on rational numbers with major errors
Ability to determine the reciprocal of real numbers and apply rational numbers in real life situations	The learner determines the reciprocals of real numbers and applies rational numbers to real life situations with varied examples	The learner determines the reciprocals of real numbers and applies rational numbers to real life situations	The learner determines the reciprocals of real numbers or applies rational numbers to real life situations with minor errors	The learner determines the reciprocals of real numbers or applies rational numbers to real life situations with minor errors
Ability to express numbers in index form, deduce the four laws and apply the laws of indices	The learner expresses numbers in index form, deduces the four laws and applies the laws of indices and gives examples	The learner expresses numbers in index form, deduces the four laws and applies the laws of indices	The learner does any two of the three (expresses numbers in index form, deduces the four laws or applies the laws of indices)	The learner does any one of the three (expresses numbers in index form, deduces the four laws or applies the laws of indices)
Ability to form quadratic expressions and equations, factorise quadratic expressions and solve quadratic equations by factorisation	The learner forms quadratic expressions and equations, factorises quadratic expressions and solves quadratic equations by factorisation, and explains each step with varied examples.	The learner forms quadratic expressions and equations, factorises quadratic expressions and solves quadratic equations by factorisation.	The learner does any two of the four (forms quadratic expressions and equations, factorises quadratic expressions or solves quadratic equations by factorisation)	The learner does any one of the four (forms quadratic expressions and equations, factorises quadratic expressions or solves quadratic equations by factorisation)

STRAND 2.0: MEASUREMENTS AND GEOMETRY

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question (s)
2.0 Measurement and Geometry	**2.1 Similarity and Enlargement** • *Similarity between objects* • *Enlargement Scale Factors (Linear, Area and Volume) (Negative scale factor will not be included)* • *Application of Similarity and Enlargement*	By the end of the sub-strand the learner should be able to: a) identify properties of similar figures, b) determine the centre of enlargement and the linear scale factor of similar figures, c) draw the image of an object given the centre of enlargement and the linear scale factor, d) determine the area scale factor of similar figures, e) determine the volume scale factor of similar objects, f) relate linear scale factor to area and volume scale factor of similar objects, g) apply similarity and enlargement to real-life situations, h) appreciate the use of similarity and enlargement in real-life situations.	The learner is guided to: • collect objects and containers from the immediate environment. discuss with peers and sort the objects and recognise those that are similar, • work with peers to observe and recognise the properties of similar objects or containers, • trace an object and its image on plain paper to locate the centre of enlargement, • measure the dimensions of the corresponding sides of an object and its image and work out its linear scale factor, • trace the object on a plain paper and use the centre and the linear scale factor to draw the image, • use area of an object and its image to work out the ratio of the image area to object area to establish the area scale factor, • use area of an object and its image to work out the ratio of the image area to object area to establish the area scale factor, • work out the volume of two similar solids such as cubes or cuboids of different sizes and establish the ratio of their volumes, • discuss with peers and establish the relationship between the linear scale factor, area scale factor and volume scale factor, • use digital devices to manipulate objects or figures to zoom or shrink sizes and use other resources to learn more on the application of similarity and enlargement in real life, **Project** Use the knowledge and skills of similarity and enlargement to make similar containers in the shape of cubes, cuboids and cylinders that are of different sizes. Identify required resources and preferably use locally available materials. Document the processes of the project including steps taken in measurements of lengths and challenges faced while doing the project. Take pictures or video clips of the project and save files in an appropriate digital device. The containers can be stored and used as a learning material.	How is similarity and enlargement applied in real life?

Core Competencies to be developed:

- Digital Literacy: The learner develops the skills of connecting and interacting with digital technology when using digital resources to learn more on the use and application of similarity and enlargement.
- Creativity and Imagination: the learner acquires networking skills when drawing images of given objects under a particular enlargement scale factor.

Values:

Responsibility: The learner takes care of objects collected from the immediate environment.

Pertinent and Contemporary Issues (PCIs):

Environment and technology: the learner acquires the skill of sorting in managing the environment after collecting and sorting the objects and containers.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement and Geometry	**2.2 Reflection** • *Symmetry and Reflection* • *Reflection on a plane surface* • *reflection on a Cartesian plane (mirror line to be x=0, y=0, y=x),*	By the end of the sub-strand, the learner should be able to: a) identify lines of symmetry in plane figures, b) determine the properties of reflection using objects and its image, c) draw an image given an object and a mirror line on a plane surface, d) draw an image given an object and a mirror line on a Cartesian plane, e) draw the mirror line given an object and its image on a plane surface and a Cartesian plane f) Apply reflection in real-life situations. g) Appreciate the role of reflection in real life situations.	The learner is guided to: • collect objects such as, sheets of paper, leaves. and identify the lines of symmetry, • work with peers to make cut outs of plane figures such as rectangles, squares and use the shapes to find the number of lines of symmetry, • stand at different distances from the plane mirror and watch the image formed. Compare the distance of the image from the mirror with the distance from the mirror and discuss the properties of reflection using the image formed, • use an object and a mirror line on plain paper to create an image to establish the properties of reflection, • use shapes and a mirror line on a plain paper to construct images formed after reflection in the mirror lines, • use shapes and mirror lines on a Cartesian plane to locate the image after reflection in the lines x=0, y=0, and y=x, • construct a mirror line given an object and its image on a plain and Cartesian plane, • discuss with peers the uses of reflection in real life such as use of driving mirrors in reducing road crashes.	How is reflection important in day-to-day activities?

Core Competencies to be developed:

- Learning to Learn: the learner develops relationship skills when discussing with peers the applications of reflections in real life such as use of mirrors in road transport.
- Creativity and Imagination: the learner acquires connection skills when making cut outs of plane figures for finding the number of lines of symmetry.

Values:

- Unity: the learner takes turns when working with peers to make cut outs of plane figures used reflection.
- Responsibility: the learn takes care of instruments used to construct images formed after reflection in the mirror lines.

Pertinent and Contemporary Issues (PCIs):

Road safety: the learner discusses with peers the applications of reflections in real life such as use of mirrors in road transport.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement and Geometry	**2.3 Trigonometry** • *Sines and Cosines of complementary angles* • *Application of trigonometric ratios (angle of elevation and depression)*	By the end of the sub strand, the learner should be able to: a) identify the sides of a right-angled triangle in relation to a particular angle, b) determine the sine, cosine and tangent of acute angles in a right-angled triangle, c) relate sines and cosines of complementary angles, d) apply trigonometric ratios to angles of elevation and depression, e) recognise the use of trigonometry in real-life situations.	The learner is guided to: • work with peers to recognise right-angled triangle in areas such doors, windows, frames and other shapes. Point out the adjacent side, opposite side to the angle as well as the hypotenuse with reference to a specific angle, • work out the sine, cosine and tangent ratios of acute angles from a right-angled triangle, • Generate a table of complementary angles. Find their sines and cosines and make conclusions, • use a simple clinometer to measure the angle of elevation and depression of objects in the immediate environment, • use trigonometric ratios to work out tasks involving angles of elevation and depression, • use digital devices and other resources to learn more about trigonometry. **Project** Make a simple clinometer using locally available materials and use it as a learning resource for measuring angles of elevation and depression.	Why is trigonometry important?

Core Competencies to be developed:

- Self-efficacy: The learner makes observations and establishes the relationships between the sine and cosine ratios.
- Learning to learn: The learner learns collaboratively while making a simple clinometer using locally available materials and using it as a learning resource.

Values:

Love: The learner demonstrates a caring attitude to others when using a simple clinometer to measure the angle of elevation and depression of different objects in the immediate environment.

Pertinent and Contemporary Issues (PCIs):

Self-esteem: The learners acquire self-esteem while making a simple clinometer using locally available materials and use it as a learning resource.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement and Geometry	**2.4 Area of Polygons** • *Area of triangle* • *Area of rhombus and parallelogram.* • *Area of a pentagon and hexagon.*	By the end of the sub-strand the learner should be able to: a) compute area of a triangle given two sides and an included acute angle, b) determine the area of a triangle given three sides using Heron's Formula, c) determine the area of a rhombus and parallelogram in different situations, d) determine the area of regular pentagon and hexagon from real life situations, e) appreciate the application of areas of polygons in real-life situations.	The learner is guided to: • identify objects with the shapes of a triangle from patterns, drawings in items such as mats, frames floor tiles. measure two sides of a triangle and the angle bounded by the two sides, use the measurements to work out the area. • work with peers to identify triangular shapes from the immediate environment such as tops of tripod stands, measure the sides and work out the area using the Heron's Formula, • identify objects with the shapes of a rhombus and parallelograms such as patterns in floors fabrics, measure the sides and an included angle and use the measurement to work out the area, • work with peers to select objects with the shape of a regular pentagons and hexagon, measure the sides and work out the area. • discuss the applications of area of polygons in day-to-day life.	How are areas of polygons useful?

Core competencies to be developed:

- Creativity and Imagination: the learner acquires networking skills by undertaking group activities and exchanging new ideas while discussing discussing the applications of area of polygons.
- Learning to learn: the learner learns independently when measuring the sides of a tringle and works out the area using the Heron's Formula.

Values:

- Unity: the learner shares available resources amicably while using selected objects with the shape of a regular pentagons and hexagon when working out the area.
- Responsibility: the learner offers leadership to others while discussing with peers the applications of area of polygons

Pertinent and Contemporary Issues (PCIs):

Social Awareness Skills: the learner develops effective communication skills when discussing with peers the applications of area of polygons.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement and Geometry	**2.5 Area of a Part of a Circle** • *Area of a sector* • *Area of a segment*	By the end of the sub-strand the learner should be able to: a) determine the area of a sector of a circle in different situations, b) determine the area of a segment of a circle in different situations, c) apply the area of part of a circle in real-life situations, d) appreciate the application of area of part of circular objects in real-life situations.	The learner is guided to: • work with peers to make cut outs of sectors of a circle, measure the radius and included angle to work out the area, • use cut outs of segments of a circle of known radius and subtended angle to work out the area of the segments, • work with peers to identify objects with the shapes of parts of a circle in the immediate environment such as area swept by an opening door on the ground, measure the radius and the angle subtended at the centre to work out the area, • discuss the applications of area part of a circle in areas such as designing patterns, cutting fruits such as water melons and decorations on different surfaces.	Where do we use area of part of a circle in real life?

Core competencies to be developed:

- Critical Thinking and Problem Solving: the learner explores different options of improvising cutouts of sectors of a circle for working out their areas.
- Learning to Learn: the learner acquires new knowledge as they discuss with peers how to work out areas of circular objects.
- Citizenship: learners appreciate their immediate environment as they gather locally available circular objects

Values:

- Unity: The learner exhibits devotion and honesty by undertaking the task while working with peers to make paper cutouts of sectors of a circle and obtain the measurements.
- Peace: the learner displays care and tolerance while discussing with peers and obtaining circular objects, tracing and marking out sectors and segments.

Pertinent and Contemporary Issues (PCIs):

Education for Sustainable Development: the learner uses locally available circular objects to trace and mark out sectors and segments, and work out the areas.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurement and Geometry	**2.6. Surface Area of Solids** • *Surface area cone, pyramid (square or rectangular base), sphere, frustum* • *Applications on area of solids.*	By the end of the sub-strand, the learner should be able to: a) determine the surface area of cones and pyramids from the nets, b) calculate the surface area of a sphere and a hemisphere, c) determine the surface area of frustums of cones and pyramids, d) apply the surfaces area of cones, pyramids and frustums in real life situations, e) appreciate the application of surface area of solids.	The learner is guided to: • works with peers to identify objects with shapes of pyramids cones and frustums such as buckets, lamp shades and tumbles to identify the shapes taken by the objects, • make models of cones and pyramids and use the nets of the models to work out the area, • use spherical objects such as soccer balls, marbles, oranges to estimate the diameter work out the area of the spheres and hemispheres, • work with peers to frustums of cones and pyramids both open and closed, measure their dimensions and work out the surface areas, • discuss the how cones, pyramids and frustums are used in day to day life.	How solids used in real life?

Core competencies to be developed:

- Critical Thinking and Problem Solving: the learner uses a cylinder and a cone of the same radius and height to relate volume of a cove to the volume of a cylinder.
- Creativity and Imagination: The learner work with peers to model frustums of cones and pyramids both open and closed.

Values:

- Social Justice: the learner accords equal opportunities when discussing the uses of pyramids and frustums are used
- Responsibility: the learner observes safety precautions of others while using locally available materials to model frustums of cones and pyramids.

Pertinent and Contemporary Issues (PCIs):

Environment and technology: The learner collects and uses locally available materials to model frustums of cones and pyramids for use as a learning resources.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurements and Geometry	**2.7 Volume and Capacity** • *Volume and capacity (cone, pyramid, frustum)* • *Application of volumes of pyramids and frustums.*	By the end of the sub-strand, the learner should be able to: a) determine the volume and capacity of cones and pyramids, b) compute volume and capacity of a frustum of a cone, c) calculate the volume and capacity of a frustum of a pyramid, d) apply volume and capacity of in real-life situations, e) appreciate volumes and capacities in real life situations.	The learner is guided to: • use model of a cone and cylinder of the same base area and height establish the relationship between volume of a cone and volume of a cylinder, • work with peers to take measurements of different containers in the shape of frustums of a cone such as bucket to work out the volume and capacity, • use model of frustum of pyramid to work out the volume and capacity, • discuss appropriate formulas and use them to work out volume and capacity of containers in the shapes of cones, pyramids and frustums. **Project** Make models of containers of different sizes in the shapes of frustums, pyramids and cones to be used as learning resources for Math. Take pictures and video clips of the process for documentation.	Why is the knowledge of volume and capacity useful?

Core competencies to be developed:

- Critical Thinking and Problem Solving: the learner acquires open-mindedness and creativity skills when discussing with peers and working out the volume and capacity of the containers.
- Creativity and Imagination: The learner develops connection or networking skills when working out the volumes of frustums from cones and pyramids.

Values:

- Unity: the learner displays tolerance of different opinions while working with peers when models of frustums as a learning resource.
- Respect: the learner listens to others' opinions while discussing appropriate formulas for working out volume and capacity of containers.

Pertinent and Contemporary Issues (PCIs):

Environment and technology: the learner uses appropriate containers from the local environment to work out the capacity of cones, pyramids, and their frustum.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Measurements and Geometry	**2.8 Commercial Arithmetic (I)** • *Budget* • *Discounts, and commission* • *Profit and loss* • *Conversion of different currencies*	By the end of the sub-strand, the learner should be able to: a) prepare budget to for use in clubs or societies, b) calculate discount and percentage discounts in different situations, c) determine commission and percentage commissions in different situations d) determine percentage profit and loss made in the sale of goods, e) convert currencies using exchange rates tables, f) recognise the role of commercial arithmetic in day-to-day life.	The learner is guided to; • brainstorm with peers how to raise funds for their clubs or societies in school set up, draw a budget showing how the money raised can be used, • role play shopping scenarios where discounts are given, work out the discounts and the percentage discount, • brainstorm scenarios under which commissions are paid to salespersons for sale of goods and services, discuss and work out the commission, • role play scenarios involving buying and selling to make profit or loss work out the percentage profit and loss make presentations to peers, • obtain digital or printed currency exchange rates charts to work out currency conversion (buying and selling) from Kenya Shillings to any other currencies and vice versa.	1. Why do we need a budget? 2. Why are discounts and commissions necessary in trading activities?

Core competencies to be developed:

- Citizenship: The learner develops ethical and digital citizenship skills when obtaining digital or printed currency exchange rates charts/tables, and working out currency conversion from Kenya Shillings to any other currencies
- Critical thinking and problem solving: The learner acquires interpretation and inference skills when role play shopping scenarios where discounts are given and work out the discounts and the percentage discount.

Values:

- Peace: the learner respects self and others by listening to others' opinion while brainstorming brainstorm scenarios under which commissions are paid to salespersons for sale of goods and services.
- Patriotism: the learner displays loyalty by showing love for one's own country's currency when obtaining digital or printed currency exchange rates tables and working out currency conversion from Kenya Shillings to any other currencies

Pertinent and Contemporary Issues (PCIs):

Financial literacy: the learner's financial skills are enhanced when obtaining digital or printed currency exchange rates charts/tables and working out currency conversion from Kenya Shillings to any other currencies.

Suggested Assessment Rubric

Level / Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to identify properties of similar figures, determine the centre of enlargement and the linear scale factor and draw the image of an object given the centre of enlargement and the linear scale factor	The learner accurately identifies properties of similar figures, determines the centre of enlargement, the linear scale factor and draws the image with high precision	The learner accurately identifies properties of similar figures, determines the centre of enlargement, the linear scale factor and draws the image	The learner does any three of the four (identifies properties of similar figures, determines the centre of enlargement, the linear scale factor and draws the image)	The learner does any two of the four (identifies properties of similar figures, determines the centre of enlargement, the linear scale factor and draws the image)
Ability to determine the area scale, determine the volume scale factor and relate linear scale factor to area and volume scale factor	The learner determines the area scale, determines the volume scale factor and relates linear scale factor to area and volume scale factor, giving examples in real life situations	The learner determines the area scale, determines the volume scale factor and relates linear scale factor to area and volume scale factor	The learner does any two of the three (determines the area scale, determines the volume scale factor and relates linear scale factor to area and volume scale factor)	The learner does any one of the three (determines the area scale, determines the volume scale factor and relates linear scale factor to area and volume scale factor)

Level / Indicators	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to identify lines of symmetry and determine the properties of reflection	The learner identifies lines of symmetry and determines the properties of reflection using illustrations	The learner identifies lines of symmetry and determines the properties of reflection	The learner identifies lines of symmetry or determines the properties of reflection with minor errors	The learner identifies lines of symmetry or determines the properties of reflection with major errors
Ability to draw an image given an object and a mirror line on a plane surface and Cartesian plane, and draw the mirror line given an object and its image	The learner draws an image given an object and a mirror line on a plane surface and Cartesian plane, and draws the mirror line given an object and its image using illustrations	The learner draws an image given an object and a mirror line on a plane surface and Cartesian plane, and draws the mirror line given an object and its image	The learner does any three of the four (draws an image given an object and a mirror line on a plane surface and Cartesian plane, and draws the mirror line given an object and its image)	The learner does any two of the four (draws an image given an object and a mirror line on a plane surface and Cartesian plane, and draws the mirror line given an object and its image)
Ability to determine the sine, cosine and tangent of acute angles	The learner accurately determines the sine, cosine and tangent of acute angles using varied examples	The learner accurately determines the sine, cosine and tangent of acute angles	The learner determines any two of the three (sine, cosine and tangent of acute angles)	The learner determines any one of the three (the sine, cosine and tangent of acute angles)
Ability to relate sines and cosines of complementary angles, apply trigonometric ratios to angles of elevation and depression	The learner correctly relates sines and cosines of complementary angles and applies trigonometric ratios to angles of elevation and depression with illustrations	The learner correctly relates sines and cosines of complementary angles and applies trigonometric ratios to angles of elevation and depression.	The learner relates sines and cosines of complementary angles or applies trigonometric ratios to angles of elevation and depression	The learner relates sines and cosines of complementary angles
Ability to compute area of a triangle given two sides and an included acute angle and determine the area of a triangle given three sides using Heron's Formula	The learner accurately computes area of a triangle given two sides and an included acute angle and determines the area of a triangle given three sides using Heron's Formula with precision	The learner accurately computes area of a triangle given two sides and an included acute angle and determines the area of a triangle given three sides using Heron's Formula	The learner computes area of a triangle given two sides and an included acute angle or determines the area of a triangle given three sides using Heron's Formula	The learner computes area of a triangle given two sides and an included acute angle
Ability to determine the area of a rhombus, parallelogram, regular pentagon and hexagon	The learner accurately determines the area of a rhombus, parallelogram, regular pentagon and hexagon in varied contexts	The learner accurately determines the area of a rhombus, parallelogram, regular pentagon and hexagon	The learner determines any three of the four (the area of a rhombus, parallelogram, regular pentagon and hexagon)	The learner determines any two of the four (the area of a rhombus, parallelogram, regular pentagon and hexagon)
Ability to determine the area of a sector, area of segment and apply the area of part of a circle	The learner determines the area of a sector, area of segment and applies the area of part of a circle in varied contexts	The learner determines the area of a sector, area of segment and applies the area of part of a circle	The learner does any two of the three (determines the area of a sector, area of segment and applies the area of part of a circle)	The learner does any one of the three (determines the area of a sector, area of segment and applies the area of part of a circle)
Ability to determine the surface area of cones, pyramids, sphere, hemisphere, frustums of cones and pyramids	The learner accurately determines the surface area of cones, pyramids, sphere, hemisphere, frustums of cones and pyramids using various examples	The learner accurately determines the surface area of cones, pyramids, sphere, hemisphere, frustums of cones and pyramids	The learner does any four of the six (determines the surface area of cones, pyramids, sphere, hemisphere, frustums of cones and pyramids)	The learner does any three of the six (determines the surface area of cones, pyramids, sphere, hemisphere, frustums of cones and pyramids)
Ability to determine the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid	The learner accurately determines the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid using various examples	The learner accurately determines the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid	The learner does any five of the eight (determines the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid)	The learner does any four of the eight (the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid)
Ability to prepare budget, calculate discount, percentage discounts, commission, percentage commissions, percentage profit and loss	The learner accurately prepares budget, calculates discount, percentage discounts, commission, percentage commissions, percentage profit and loss in varied contexts	The learner accurately prepares budget, calculates discount, percentage discounts, commission, percentage commissions, percentage profit and loss	The learner does any four of (prepares budget, calculates discount, percentage discounts, commission, percentage commissions, percentage profit and loss)	The learner does any three of (prepares budget, calculates discount, percentage discounts, commission, percentage commissions, percentage profit and loss)

STRAND 3.0: STATISTICS AND PROBABILITY

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Statistics and Probability	**3.1 Statistics I** • *Ungrouped data* • *Application of graphs of data*	By the end of the sub-strand the learner should be able to: a) draw a frequency distribution table for ungrouped data b) determine mean, mode and median of ungrouped data, c) represent data using bar graphs, line graphs and pie charts, d) interpret data from bar graphs, line graphs and pie charts, e) appreciate the importance of data organization and interpretation in making effective decisions	The learner is guided to: • collect data from the locality such as means of transport used by learners to school, types of trees in the school compound and generate a frequency distribution table for discrete data, • collect data from areas such as the most favourite colour for learners to work out the mean, mode, median for discrete data, • use collected from the school environment to draw bar graph, line graph and pie chart, • work with peers to extract information from graphs and charts used to represent the data and draw conclusions, • work with peers to practise representing data collected using bar graph, line graph and pie charts on a digital device. **Project** Identify a situation in the school or community. Collect data on it, organise the data and represent it using graphs and charts. Interpret the graphs and charts and use the information to guide the school or community to make informed decisions.	How do we use statistics in day-to-day life?

Core Competencies to be developed:

- Critical thinking and problem solving: the learner develops interpretation and inference skills when collecting data.
- Digital literacy: the learner develops the skills of connecting and interacting with digital technology when using digital devices to draw graphs and charts to represent the data collected.

Values:

- Integrity: the learner displays honesty while working in groups and collecting data such making frequency distribution tables.
- Love: the learner is considerate of the other's when working with peers to extract information from graphs and charts draw conclusions

Pertinent and Contemporary Issues (PCIs):

Life Skills (Decision Making): the learner makes informed decisions when identifying patterns from the graphs and charts and drawing conclusions.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question (s)
3.0 Statistics and Probability	**4.2 Probability 1** • *Equally and likely events* • *Mutually exclusive events* • *Independent events (maximum of two experiments and two outcomes)*	By the end of the sub-strand, the learner should be able to: a) perform experiments involving equally likely outcomes in different situations, b) compute probability of equally likely outcomes, c) determine the range of probability of an event, d) identify mutually exclusive events in real life situations, e) perform experiments of single chance involving mutually exclusive events, f) calculate the probability of an event involving a single chance, g) perform experiments involving independent events, h) determine the probability of independent events in different situations, i) appreciate the role of probability in real-life situations.	The learner is guided to: • carry out experiments of events involving equally likely outcomes such as tossing of a coin, • discuss with peers and work out the probability of equally likely events such as rolling a six faced dice, • discuss the range of probability using real life scenarios, • work with peers to recognise events that are mutually exclusive, • carry out experiments involving mutually exclusive events to work out the probability possible outcomes, • carry out experiments involving independent events and work out the probability of the possible outcomes, • discuss the dangers of gambling using the knowledge of probability, • use digital or other resources to explore more on probability	How do we apply probability in day to day life?

- Communication and collaboration: the learner evaluating successes and failures from experiments involving mutually exclusive events to work out the probability possible outcomes.
- Critical thinking and problem solving: the learner develops interpretation and inference when discussing the dangers of gambling using the knowledge of probability.

Values:

- Responsibility: The learner makes correct decisions on life issues that involves gambling using the knowledge of probability.
- Unity. The learner collaborates with others when carry out experiments involving probability.

Pertinent and Contemporary Issues (PCIs):

Life Skills (Decision Making): The learner makes informed decisions on life issues involving probability such dangers of gambling.

Suggested Assessment Rubric

Level / Indicators	Exceeds Expectations	Meets Expectations	Approaches expectations	Below Expectations
Ability to determine mean, mode, and median of ungrouped data	The learner correctly works out mean, mode, and median of ungrouped data using various examples	The learner accurately works out mean, mode, and median of ungrouped data	The learner works out any two of (mean, mode and median of ungrouped data)	The learner works out any one of (mean, mode and median of ungrouped data)
Ability to represent data using graphs and charts	The learner creatively and correctly represents data using graphs and charts	The learner correctly represents data using graphs and charts	The learner represents data using graphs or charts with minor errors	The learner represents data using graphs or charts with major errors
Ability to interpret data from graphs and charts	The learner accurately and comprehensively interprets data from charts and graphs	The learner accurately interprets data from charts and graphs	The learner interprets data from charts or graphs with minor errors	The learner interprets data from charts or graphs with major errors
Ability to perform experiments involving equally likely, mutually exclusive events and independent events	The learner performs experiments involving equally likely, mutually exclusive events and independent events in varied contexts	The learner performs experiments involving equally likely, mutually exclusive events and independent events	The learner performs experiments involving any two of the three (equally likely, mutually exclusive events and independent events)	The learner performs experiments involving any one of the three (equally likely, mutually exclusive events and independent events)
Ability to compute probability of equally likely events, range of probability, single outcome, and probability of independent events	The learner computes probability of equally likely events, range of probability, single outcome, and probability of independent events in varied contexts	The learner computes probability of equally likely events, range of probability, single outcome, and probability of independent events	The learner computes any two of the three (probability of equally likely events, range of probability, single outcome, and probability of independent events)	The learner computes any one of the three (probability of equally likely events, range of probability, single outcome, and probability of independent events)

APPENDIX 1: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
1.0: Numbers and Algebra	1.1: Real Numbers	• Class activities • Class written tests • portfolio	Games on charts, number cards.	Clubs
	1.2: Indices and Logarithms	• Class activities • Class written tests	Mathematical tables Calculators	Clubs
	1.3: Quadratic Equations	• Class activities • Class written tests • portfolio	Rulers, drawing tools, graph papers/squared books	Clubs
2.0: Measurement and Geometry	2.1: Similarity and Enlargement	• Class activities • Class written tests • Projects • Rating scale • Observation schedule	Rulers, drawing tools, geometrical set, calculators, objects with shapes of different polygons, triangular cut outs	Clubs
	2.2: Reflection	• Class activities • Class written tests • Projects • Rating scale • Observation schedule	Rulers, drawing tools, geometrical set, calculators, objects with shapes of different polygons, circular cut outs	Clubs

Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
	2.3: Trigonometry	• Class activities • Class written tests	Rulers, drawing tools, graph papers/ squared books, geometrical set, models of solids, cutting tools	Clubs
	2.4: Area of Polygons	• Class activities • Class written tests • Projects • portfolio	Rulers, drawing tools, graph papers/ squared books, geometrical set, models of solids, cutting tools	Clubs
	2.5: Area of Part of a Circle	• Class activities • Class written tests	Drawing tools rulers, , graph papers/ squared books	Clubs
	2.6 Surface Area of Solids	• Class activities • Class written tests • Portfolio • Rating scale • Observation schedule	Rulers, drawing tools, graph papers/ squared books, geometrical set, models of similar figures and shapes	Clubs
	2.7 Volume and Capacity	• Class activities • Class written tests • Projects	Rulers, drawing tools, graph papers/ squared books, geometrical set, containers.	Clubs

Strand	Sub-strand	Suggested Assessment Methods	Suggested Learning Resources	Suggested Non-Formal Activities
	2.8 Commercial Arithmetic 1	• Class activities • Class written tests • Rating scale • Observation schedule	Currency conversion tables, improvised currencies/dummies , Rulers, drawing tools, graph papers/ squared books, geometrical set.	Clubs
3.0: Statistics and Probability	3.1 Statistics I	• Class activities • Class written tests • Rating scale • Observation schedule	Data from different sources, rulers, drawing tools, graph papers/ squared books, geometrical set, calculators	Clubs

APPENDIX 2: USE OF ICT TOOLS

The following ICT tools may be used in learning and teaching of Mathematics at this level:

1. Learner digital devices (LDD)
2. Teacher digital devices (TDD)
3. Mobile phones
4. Digital clocks
5. Television sets
6. Videos
7. Cameras
8. Projectors
9. Radios
10. DVD players
11. CDs
12. Scanners
13. Internet
14. Other relevant digital resources e.g. Online Education Resources such as the Kenya Education Cloud.